

Cloud access*

Distributed Data Processing Environments

Lab Guide 3

This session aims at creating and disposing of cloud instances. Set up your account, your budget, and your quotas as described in the cloud protocol before starting.

Steps First create an instance from the Cloud Console:

1. Create a new instance selecting:
 - region *us-central1* and zone *us-central1-a*;
 - machine type: e2-micro;
 - operating system type Ubuntu;
 - version 24.04 LTS (x86_64).
2. Before creating it, note the estimated monthly cost.
3. Change the machine type to e2-medium, and then to e2-standard-2, noting the estimate each time. Change the region to *europa-west1* and note it again.
4. Restore the original selection, choose “Create”, and wait for the instance to become ready.

Now connect to the instance and work in it:

1. Connect with `gcloud compute ssh`.
2. Install packages and run the program from Lab 1. Copy files with `gcloud compute scp`.
3. Ask the agent what this instance costs per month. Compare its answer with the estimate you noted above, and account for every line item that it omitted.
4. Delete the instance from the console.

Finally, replace all of the above with a description:

1. Create an empty folder and copy into it the `main.tf` from Lab 2.
2. Replace the Docker provider with the Google provider:

```
terraform {
  required_providers {
    google = { source = "hashicorp/google", version = "~> 6.0" }
  }
}

provider "google" {
  project = var.project
  zone    = "us-central1-a"
}
```

*Adversary mode in the second part. See AI protocol.

It authenticates with the credentials left by `gcloud auth application-default login`.

3. Replace `docker_container` with a `google_compute_instance`,¹ carrying the zone, machine type, and image you selected by hand above.
4. Add a `scheduling` block so that the instance deletes itself after 8 hours.
5. Predict what `tofu plan` will report, then run it, and then apply.
6. Connect to the instance and confirm that it is the one you described.
7. Change the machine type in the file, and explain whether `plan` updates the instance or replaces it.

At the end of the session, run `tofu destroy` and follow the teardown checklist in the cloud protocol.

Learning Outcomes Create and dispose of cloud instances, by hand and from a description. Relate instance type and location with cost.

¹https://registry.terraform.io/providers/hashicorp/google/latest/docs/resources/compute_instance